

# **STATSIG lecture 3**

**Analysis of random signals in frequency  
domain: power spectral density  
Filtering of random signals**

## **Disclaimer:**

**Please note that these slides are only a brief overview of the topic.  
Many more details will be explained on the blackboard and with  
MATLAB examples**

# Power spectral density, PSD

Description of random signal  $X(t)$  in frequency domain

## Definition

$$S_{xx}(\omega) \stackrel{\text{def}}{=} \mathcal{F}\{R_{xx}(\tau)\} = \int_{-\infty}^{\infty} R_{xx}(\tau) e^{-j\omega\tau} d\tau \quad \left[\frac{V^2}{Hz}\right]$$

## Power in bandwidth $\omega \in [\omega_1, \omega_2]$

$$P_{[\omega_1, \omega_2]} = 2 \frac{1}{2\pi} \int_{\omega_1}^{\omega_2} S_{xx}(\omega) d\omega \quad [V^2]$$

## Total power

$$P_{total} = E[X^2(t)] = R_{xx}(0) = 2 \frac{1}{2\pi} \int_0^{\infty} S_{xx}(\omega) d\omega \quad [V^2]$$

## Properties

$$\begin{aligned} S_{xx}(\omega) &\text{ real} \\ S_{xx}(\omega) &\geq 0 \\ S_{xx}(-\omega) &= S_{xx}(\omega) \end{aligned}$$

## Cross power spectral density

$$S_{xy}(\omega) \stackrel{\text{def}}{=} \mathcal{F}\{R_{xy}(\tau)\} = \int_{-\infty}^{\infty} R_{xy}(\tau) e^{-j\omega\tau} d\tau$$

## Properties

$$\begin{aligned} S_{xy}(\omega) &\text{ complex} \\ S_{xy}(-\omega) &= S_{xy}^*(\omega) \end{aligned}$$

## Bandwidth of random signal X(t)

LP signal (frequency content centered around DC, baseband signal)

Noise equivalent (effective) bandwidth  $B_{eff} = \frac{\int_0^\infty S_{xx}(\omega) d\omega}{S_{xx}(0)}$

BP signal (frequency content centered around carrier frequency  $\omega_0$ , bandpass signal)

Noise equivalent (effective) bandwidth  $B_{eff} = \frac{\int_0^\infty S_{xx}(\omega) d\omega}{S_{xx}(\omega_0)}$

# Filtering of WSS random signal $X(t)$ to WSS random signal $Y(t)$

LTI-filter described by impulseresponse,  $h(t)$  and frequency response function,  $H(f)$

As always

$$Y(t) = h(t) * X(t)$$
$$Y(f) = H(f)X(f)$$

Stochastic properties

$$\mu_Y = H(0)\mu_X$$
$$R_{xy}(\tau) = h(\tau) * R_{xx}(\tau)$$
$$R_{yx}(\tau) = h(-\tau) * R_{xx}(\tau)$$
$$R_{yy}(\tau) = h(\tau) * h(-\tau) * R_{xx}(\tau)$$
$$S_{xy}(\omega) = H(\omega)S_{xx}(\omega)$$
$$S_{yy}(\omega) = |H(\omega)|^2 S_{xx}(\omega)$$
$$P_y = \sigma_Y^2 = R_{yy}(0) = 2 \frac{1}{2\pi} \int_0^\infty S_{yy}(\omega) d\omega = 2 \frac{1}{2\pi} \int_0^\infty |H(\omega)|^2 S_{xx}(\omega) d\omega$$

Often occuring special case:  $X(t)$  is zero-mean white noise

$$\mu_X = 0$$
$$S_{xx}(\omega) = \sigma_X^2$$
$$R_{xx}(\tau) = \sigma_X^2 \delta(\tau)$$

## Effective bandwidth of filter

LP filter       $B_{eff} = \frac{\int_0^\infty |H(\omega)|^2 d\omega}{|H(0)|^2}$

BP filter       $B_{eff} = \frac{\int_0^\infty |H(\omega)|^2 d\omega}{|H(\omega_0)|^2}$

## Coherence function for random signals X(t), Y(t)

Measure of to what degree (0-100%) random signal Y(t) can be explained as linear filtering of random signal X(t)

**Definition**  $\gamma_{xy}^2(\omega) \stackrel{\text{def}}{=} \frac{|S_{xy}(\omega)|^2}{S_{xx}(\omega)S_{yy}(\omega)} \quad 0 \leq \gamma_{xy}^2(\omega) \leq 1$

Compare with correlation coefficient between random variables X and Y:  $\rho_{xy} \stackrel{\text{def}}{=} \frac{\sigma_{xy}}{\sigma_x \sigma_y}$

Measures to what degree (0-100%) random variable Y is a linear combination of X

**Pure linear filtering**  $\gamma_{xy}^2(\omega) = 1$

If  $\gamma_{xy}^2(\omega) < 1$       **non-linear filter**  
noisy measurement of X(t) and/or Y(t)  
more inputs than X(t)